Madison **Hanberry**

SOFTWARE ENGINEER · CLOUD AUTOMATION EXPERT · PHYSICS RESEARCHER

Seattle WA, USA

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Skills

Cloud Platforms	Amazon AWS, Microsoft Azure, Google Cloud, Digital Ocean
Infrastructure Automation	Terraform, Ansible, Docker, Kubernetes, VMware vSphere, vRealize Automation, vRealize Orchestrator, NSX Firewall, F5 Load Balancers, BlueCat DNS, Enterprise and Non-Enterprise GNU/Linux Systems, AIX
Programming Languages	Python, C/C++, JAVA, Node.js/Javascript, PHP, OCaml, LaTeX, Bash, Fortran, Matlab, ARM/NASM Assembly
Web Technologies	HTML, CSS, jQuery, angular.js, react.js, modular.js, CGI web backend implementation, REST API implementation
Database Systems	SQL, REDIS, MongoDB

Experience

TriMagnetix — *Seattle, Washington*

CO-FOUNDER, CHIEF OF R&D — *August 2025 - Present*

- Oversees technical strategy and design for nanomagnetic logic chips
- Implements chip prototypes in a level-5 clean room at the Washington Nanofabrication Lab
- Leads a team of engineers in creating nanomagnetic logic design and simulation tools
- Maintains stakeholder and investor relationships for the organization

Amazon — *Seattle, Washington*

SOFTWARE DEV ENGINEER II — *December 2023 - August 2025*

- Designed and implemented a platform to accelerate Amazon's business expansions
- Achieved an 80% reduction in operation cost and a 30% increase in velocity of expansion operations
- Designed and implemented a platform to optimize and redistribute teams based on skill level and location
- Reduced team reorganization lead time from months to weeks

Very Good Security — *Seattle, Washington*

SENIOR SOFTWARE ENGINEER — *March 2022 - December 2023*

- Implemented and brought to market a file processing and obfuscation product for enterprise customers
- Implemented a standardized approach to IaC and CI/CD with company-wide impact
- Overhauled the company's observabiliy stack for 60% cost savings and increased reliability

Home Depot — *Seattle, Washington*

SENIOR SITE RELIABILITY ENGINEER — *March 2021 - March 2022*

- Maintained Google Cloud Infrastructure and Policies
- Implemented IaC for production and non-production environments using Terraform
- Implemented continuous integration pipelines using Circle CI
- Implemented continuous deployment pipelines to Google Kubernetes Engine clusters using Harness and Spinnaker

Symetra — *Seattle, Washington*

SENIOR FULL STACK ENGINEER — *September 2020 - March 2021*

- Created an automated insurance approval system
- Built an entire platform on AWS technologies utilizing Serverless
- Implemented a machine learning to aid in the evaluation of insurance candidates
- Developed a distributed blockchain ledger to track contract history

AppyMeal
Seattle, WA & Atlanta, GA

Co-Founder & Lead Software Engineer
April 2019 - PRESENT

- Designed and implemented everything in the AppyMeal app (frontend, backend, payment processing, identity management, PCI compliance, etc.)
- Automated the server infrastructure for hands-off maintenance and lean operation
- Led a team of 6 developers and designers
- Successfully Launched on Google Play and Apple App Store

AIM Consulting
Seattle, Washington

Cloud Engineering and Software Development Consultant
June 2020 - September

- Worked on creating a FHIR-compliant REST api for Medinformatix
- Built serverless endpoints on AWS Lambda
- Implemented a go-forward CI/CD solution built on Cloudformation, CodeBuild, and Codepipeline

Fiserv
Atlanta, Georgia

Cloud Automation Engineer
Aug. 2017 - June 2020

- Led the design and implemtation of Fiserv's hybrid-cloud platform
- Created cloud-agnostic solutions to standardize infrastructure-as-code for Azure, AWS, GCP, and vSphere
- Deployed and managed multi-cloud kubernetes clusters utilizing vanilla kubernetes, Azure Kubernetes Service, Amazon EKS, Google Kubernetes Engine, and Rancher
- Deployed and managed additional container solutions including docker swarm and pivotal cloud foundry
- Engineered standardized CI/CD solutions based on Jenkins, Azure Pipelines, and GitLab CI
- Contributed to design and delivery of the IaaS platform
- Created a self-service portal for the Fiserv Enterprise Hybrid Cloud
- Main contributor for cloud integration efforts during the Fiserv and First Data merger
- Created a chatbot from scratch to offload common support and devOps tasks

Georgia State University Center for Nano-Optics
Atlanta, Georgia

Staff Spintronics Researcher
Jan. 2015 - May 2018

- Designed a non-volitile base-six computer processor utilizing directional anisotropy in namomagnetic triangle arrays
- Designed and implemented experiment-control interfaces
- Created software for fractal dimension analysis of magnetic domains
- Automated image analysis of MOKE microscopy footage
- Designed and simulated nanomagnetic interfaces

Georgia State University Center for Excellence in Teaching and Learning
Atlanta, Georgia

Student Innovation Fellow
July 2016 - July 2017

- Engineered software solutions for research teams at collaborating universities
- The subject matter was diverse and included the following:
  - Diabetes Treatment
  - Cognitive Development
  - Political Science
  - Literature and Language Analysis

Open Source

NMAG Nanomagnetic Simulator
nmag.soton.ac.uk/nmag

Maintainer & Contributor
2017 - PRESENT

- NMAG is a nanomagnetic simulator that has been cited in over 300 publications.
- Wrote a patch in 2017 that allowed it to be compiled easily with a modern software stack on Linux
- Continued maintaining said patch in the coming years.
- The patch saw significant use and led the creator of NMAG (Hans Fongohr), to ask if I would like to become the maintainer of the project in 2019.
- Since becoming the project maintainer, I have made the following contributions:
  - Worked to port the project off of the southampton.edu servers
  - Containerized the application using the singularity container platform
  - Worked to modernize the codebase.

Education

Georgia Tech (Georgia Institute of Technology) — *Atlanta, Georgia*

M.S. in Computer Science with a focus in Computer Architecture — *2023*

- Constructed a hypervisor management daemon using libvirt
- Implemented extensive processor caching mechanisms for MIPS emulation
- Analyzed and created a patch for malware within a sandboxed environment

Georgia State University — *Atlanta, Georgia*

B.S. in Computer Science wit a focus in Theoretical Computer Science — *2018*

- Received awards for research and academic excellence
- Staff researcher in Dr. Alexander Kozhanov's spintronics research lab
- Contributed to numerous research teams accross multiple disciplines including:
  - Spintronics
  - Cancer Cell Migration
  - Diabetes Treatment
  - Cognitive Development
  - Political Science
  - Literature and Language Analysis

Awards & Certifications

Certifications

2021	**Google Cloud Architect**, GCA Exam October 2021	*Seattle, WA*
2019	**Google Cloud Engineer**, GCE Exam at Google Next 2019	*San Francisco, CA*

Awards

2016	**Best Oral Presentation**, GSURC for the presentation of *Triad Computing*	*Atlanta, Georgia*
2016	**Who's Who Among Students**, Georgia State University for academic excellence	*Atlanta, Georgia*
2014-18	**Honor Roll**, Georgia State University	*Atlanta, Georgia*

Presentations

Switching Dynamics in Triangular Nanomagnets — *New Orleans, Louisiana*

First Author & Presenter, American Physical Society March 2017 Meeting — *March 2017*

- Unveiled simulation results of complex triangular nanomagnetic systems
- Detailed how said systems could be used to implement a non-volitile base-six processor

Dzyaloshinskii-Moria Interaction in CoNiPt Tri-Layer Heterostructures — *New Orleans, Louisiana*

Secondary Author, American Physical Society March 2017 Meeting — *March 2017*

- Detailed experimental observation and analysis of the DMI effect in a CoNiPt sample

Magnetization Reversal Dynamics in CoNi Heterostructures — *New Orleans, Louisiana*

Secondary Author, American Physical Society March 2017 Meeting — *March 2017*

- Detailed experimental observation and analysis of magnetization reversal in various CoNi samples

Spin Waves Propagation in Structured Magnetic Films with Perpendicular Magnetic Anisotropy — *New Orleans, Louisiana*

Secondary Author, American Physical Society March 2017 Meeting — *March 2017*

- Detailed results and analysis of spin wave simulation in thin magnetic films
- Summarized the potential for applications in computer logic

Triad Computing — *Atlanta, Georgia*

First Author & Presenter, 2016 Georgia State University Undergraduate Research Conference — *March 2016*

- Outlined the potential for higher-base computing using novel magnetic approaches, particularly the use of nanomagnetic triangles, or *triads*
- This was awarded first place for *Best Oral Presentation*